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Micro light-emitting diode device

Abstract

A micro light-emitting diode device includes a substrate, a micro light-emitting diode, and a cathode transparent electrode. The micro light-emitting diode is disposed on the substrate and includes a p-type III-nitride layer, a first n-type III-nitride layer above the p-type III-nitride layer, a second n-type III-nitride layer above the first n-type III-nitride layer, and an active layer between the p-type and first n-type III-nitride layers. The second n-type III-nitride layer contains aluminum and has top and bottom surfaces. A refractive index of the second n-type III-nitride layer is smaller than a refractive index of the first n-type III-nitride layer and varies in a monotonically non-decreasing manner from the top surface. The refractive index of the second n-type III-nitride layer is larger at the bottom surface than at the top surface. The cathode transparent electrode is in contact with the top surface.

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Background/Summary

BACKGROUND

Field of Invention

(1) The present disclosure relates to a micro light-emitting diode device.

Description of Related Art

(2) The statements in this section merely provide background information related to the present disclosure and do not necessarily constitute prior art.

(3) As a light source, light-emitting diodes (LEDs) have many advantages, including low energy consumption, long lifetime, small size, and fast switching. Hence, conventional lighting, such as incandescent lighting, is gradually replaced by LED lights. The properties regarding LEDs also fit applications on displays. Researches on displays using micro light-emitting devices, or specifically, micro light-emitting diodes (micro LEDs), have become popular in recent years. Commercial lighting applications made of micro LEDs are nearly within reach.

(4) GaN (InGaN, AlGaN) based LEDs have become the majority of LED Illumination. Since the GaN based LED display gradually becomes mainstream in the display market, searching for better performance for the GaN based LED display becomes an important topic, and the display industries are starting to pay attention on it.

SUMMARY

(5) According to some embodiments of the present disclosure, a micro light-emitting diode device

includes a substrate, a micro light-emitting diode, and a cathode transparent electrode. The micro light-emitting diode has a lateral width smaller than 100 μm . The micro light-emitting diode is disposed on and in contact with the anode electrode. The micro light-emitting diode includes a p-type III-nitride layer, a first n-type III-nitride layer, a second n-type III-nitride layer, and an active layer. The first n-type III-nitride layer is above the p-type III-nitride layer. The second n-type III-nitride layer is above and in contact with the first n-type III-nitride layer. The second n-type III-nitride layer contains aluminum and has a top surface and a bottom surface respectively away from and adjacent to the first n-type III-nitride layer. A refractive index of the second n-type III-nitride layer is smaller than a refractive index of the first n-type III-nitride layer and varies in a monotonically non-decreasing manner from the top surface. The refractive index of the second n-type III-nitride layer is larger at the bottom surface than at the top surface. The active layer is between the p-type III-nitride layer and the first n-type III-nitride layer. The cathode transparent electrode is at least partially in contact with the top surface of the second n-type III-nitride layer. (6) It is to be understood that both the foregoing general description and the following detailed description are by examples, and are intended to provide further explanation of the disclosure as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:
- (2) FIG. 1 is a schematic cross-sectional view of a micro light-emitting diode device according to some embodiments of the present disclosure;
- (3) FIG. 2A is a graph showing distribution of an atomic ratio of aluminum in a second n-type III-nitride layer in a thickness direction according to some embodiments of the present disclosure;
- (4) FIG. 2B is a graph showing the distribution of the atomic ratio of aluminum in the second n-type III-nitride layer in the thickness direction according to some embodiments of the present disclosure;
- (5) FIG. 2C is a graph showing the distribution of the atomic ratio of aluminum in the second n-type III-nitride layer in the thickness direction according to some embodiments of the present disclosure;
- (6) FIG. 2D is a graph showing the distribution of the atomic ratio of aluminum in the second n-type III-nitride layer in the thickness direction according to some embodiments of the present disclosure;
- (7) FIG. 3A is a graph showing distribution of a refractive index of the second n-type III-nitride layer of FIG. 2A in the thickness direction according to some embodiments of the present disclosure;
- (8) FIG. 3B is a graph showing distribution of the refractive index of the second n-type III-nitride layer of FIG. 2B in the thickness direction according to some embodiments of the present disclosure;
- (9) FIG. 3C is a graph showing distribution of the refractive index of the second n-type III-nitride layer of FIG. 2C in the thickness direction according to some embodiments of the present disclosure;
- (10) FIG. 3D is a graph showing distribution of the refractive index of the second n-type III-nitride layer of FIG. 2D in the thickness direction according to some embodiments of the present disclosure;
- (11) FIG. 4 is a schematic diagram showing two structures;
- (12) FIG. 5 is a partial schematic cross-sectional view of a cathode transparent electrode according

to some embodiments of the present disclosure;

(13) FIG. 6 is a schematic cross-sectional view of a micro light-emitting diode device according to some embodiments of the present disclosure;

(14) FIG. 7A is a schematic cross-sectional view of a micro light-emitting diode device according to some embodiments of the present disclosure;

(15) FIG. 7B is a schematic cross-sectional view of a micro light-emitting diode device according to some embodiments of the present disclosure;

(16) FIG. 8 is a schematic cross-sectional view of a micro light-emitting diode device according to some embodiments of the present disclosure;

(17) FIG. 9 is a schematic cross-sectional view of a micro light-emitting diode device according to some embodiments of the present disclosure;

(18) FIG. 10 is a schematic cross-sectional view of a micro light-emitting diode device according to some embodiments of the present disclosure;

(19) FIG. 11 is a schematic top view of a micro light-emitting diode device according to some embodiments of the present disclosure; and

(20) FIG. 12 is a schematic top view of a micro light-emitting diode device according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

(21) Reference will now be made in detail to the present embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the description to refer to the same or like parts.

(22) In various embodiments, description is made with reference to figures. However, certain embodiments may be practiced without one or more of these specific details, or in combination with other known methods and configurations. In the following description, numerous specific details are set forth, such as specific configurations, dimensions, and processes, etc., in order to provide a thorough understanding of the present disclosure. In other instances, well-known semiconductor processes and manufacturing techniques have not been described in particular detail in order to not unnecessarily obscure the present disclosure. Reference throughout this specification to “one embodiment,” “an embodiment”, “some embodiments” or the like means that a particular feature, structure, configuration, or characteristic described in connection with the embodiment is included in at least one embodiment of the disclosure. Thus, the appearances of the phrase “in one embodiment,” “in an embodiment”, “according to some embodiments” or the like in various places throughout this specification are not necessarily referring to the same embodiment of the disclosure. Furthermore, the particular features, structures, configurations, or characteristics may be combined in any suitable manner in one or more embodiments.

(23) The terms “micro” device, “micro” p-n diode or “micro” LED as used herein may refer to the descriptive size of certain devices or structures according to embodiments of the present disclosure. As used herein, the terms “micro” devices or structures may be meant to refer to the scale of 1 to 100 μm . However, it is to be appreciated that embodiments of the present disclosure are not necessarily so limited, and that certain aspects of the embodiments may be applicable to larger, and possibly smaller size scales. The refractive index mentioned in the following, unless otherwise specified, is based on the refractive index of light with a wavelength λ of 550 nm.

(24) Reference is made to FIG. 1. FIG. 1 is a schematic cross-sectional view of a micro light-emitting diode device 100A according to some embodiments of the present disclosure. As shown in FIG. 1, in some embodiments, the micro light-emitting diode device 100A includes a substrate 110, a micro light-emitting diode 120, and a cathode transparent electrode 130. The micro light-emitting diode 120 has a lateral width LW smaller than 100 μm . For example, the lateral width LW is the width of micro light-emitting diode 120 shown in FIG. 1. The substrate 110 has an anode electrode 111 thereon. The micro light-emitting diode 120 is disposed on and in contact with the anode electrode 111. The micro light-emitting diode 120 includes a p-type III-nitride layer 121, a first n-

type III-nitride layer **123**, a second n-type III-nitride layer **124**, and an active layer **122**. The first n-type III-nitride layer **123** is above the p-type III-nitride layer **121**. The second n-type III-nitride layer **124** is above and in contact with the first n-type III-nitride layer **123**. The second n-type III-nitride layer **124** contains aluminum and has a top surface **124a** and bottom surface **124b** respectively away from and adjacent to the first n-type III-nitride layer **123**. A refractive index of the second n-type III-nitride layer **124** is smaller than a refractive index of the first n-type III-nitride layer **123** and varies in a monotonically non-decreasing manner from the top surface **124a**. The refractive index of the second n-type III-nitride layer **124** is larger at the bottom surface **124b** than at the top surface **124a**.

(25) In some embodiments, the refractive index of the second n-type III-nitride layer **124** monotonically increases from the top surface **124a** to the bottom surface **124b** of the second n-type III-nitride layer **124**. In other words, the refractive index of the second n-type III-nitride layer **124** varies in a way that only ever increases from the top surface **124a**. That is, the refractive index of the second n-type III-nitride layer **124** has a minimum value at the top surface **124a** and a maximum value at the bottom surface **124b**, and the maximum value is still smaller than the refractive index of the first n-type III-nitride layer **123**. “III” (group III) of each of the p-type III-nitride layer **121**, the first n-type III-nitride layer **123**, and the second n-type III-nitride layer **124** refers to (or consists of) at least one element of aluminum, gallium, and indium. The active layer **122** is between the p-type III-nitride layer **121** and the first n-type III-nitride layer **123**. The cathode transparent electrode **130** at least partially covers and is in contact with the top surface **124a** of the second n-type III-nitride layer **124**.

(26) It should be pointed out that since the second n-type III-nitride layer **124** contains aluminum relative to the first n-type III-nitride layer **123**, the band gap of the second n-type III-nitride layer **124** is wider than the band gap of the first n-type III-nitride layer **123**. In general, the wider band gap of the second n-type III-nitride layer **124** causes the refractive index thereof to be smaller than the refractive index of the first n-type III-nitride layer **123** with the narrower band gap.

(27) By the way, the aluminum content in an III-nitride material has a positive influence to the band gap of the III-nitride material. The higher the atomic ratio of aluminum the material has, the wider the band gap of the material. Thus, in some embodiments, an atomic ratio of aluminum in the second n-type III-nitride layer **124** varies in a monotonically non-increasing manner from the top surface **124a**. In this way, the second n-type III-nitride layer **124** having the refractive index varying in a monotonically non-decreasing manner from the top surface **124a** can be obtained.

(28) For example, reference is made to FIGS. 2A, 2B, and 2C. Each of FIGS. 2A to 2C is a graph showing distribution of the atomic ratio of aluminum in the second n-type III-nitride layer **124** in a thickness direction according to some embodiments of the present disclosure. The thickness direction may refer to a direction perpendicular to the top surface **124a** or the bottom surface **124b**. As shown in each of FIGS. 2A, 2B, and 2C, the atomic ratio of aluminum in the second n-type III-nitride layer **124** varies in a monotonically non-increasing manner from the top surface **124a**.

(29) Reference is made to FIGS. 3A, 3B, and 3C. FIGS. 3A to 3C are graphs showing distributions of the refractive indices of the second n-type III-nitride layers **124** respectively of FIGS. 2A to 2C in the thickness direction according to some embodiments of the present disclosure. As shown in each of FIGS. 3A, 3B, and 3C, the refractive index of the second n-type III-nitride layer **124** varies in a monotonically non-decreasing manner from the top surface **124a**.

(30) Reference is made to FIG. 2D. FIG. 2D is a graph showing the distribution of the atomic ratio of aluminum in the second n-type III-nitride layer **124** in the thickness direction according to some embodiments of the present disclosure. As shown in FIG. 2D, in some embodiments, the atomic ratio of aluminum in the second n-type III-nitride layer **124** monotonically decreases from the top surface **124a** to the bottom surface **124b**. In other words, the atomic ratio of aluminum in the second n-type III-nitride layer **124** varies in a way that only ever decreases from the top surface **124a**. That is, the atomic ratio of aluminum of the second n-type III-nitride layer **124** has a

minimum value at the bottom surface **124b** and a maximum value at the top surface **124a**.

(31) Reference is made to FIG. 3D. FIG. 3D is a graph showing distribution of the refractive index of the second n-type III-nitride layer **124** of FIG. 2D in the thickness direction according to some embodiments of the present disclosure. As shown in FIG. 3D, the second n-type III-nitride layer **124** having the refractive index monotonically increases from the top surface **124a** can be obtained.

(32) In some embodiments, the second n-type III-nitride layer is formed by a deposition process, and the doping amount of aluminum can be dynamically adjusted during the deposition process.

(33) In some embodiments, a refractive index of the cathode transparent electrode **130** is smaller than the refractive index of the second n-type III-nitride layer **124** at the top surface **124a**. In other words, the refractive index of the second n-type III-nitride layer **124** is between the refractive index of the first n-type III-nitride layer **123** and the refractive index of the cathode transparent electrode **130**.

(34) According to Snell's Law, the phenomenon of refraction occurs when light propagates from one medium to another medium with a different refractive index. When the difference in refractive index between the two mediums is large, more light will be reflected at the interface between the two mediums. In detail, assuming that light travels from an optically dense medium (e.g., the first n-type III-nitride layer **123**) with a refractive index of n_1 to an optically sparser medium (e.g., the second n-type III-nitride layer **124**) with a refractive index of n_2 smaller than n_1 , and the incident angle θ_1 is equal to the critical angle θ_c , the refracted ray will proceed along the tangent of the interface. The critical angle θ_c is the smallest angle of incidence at which total internal reflection occurs and can be determined by the following equation (1).

$$\theta_c = \sin^{-1}(n_2/n_1) \quad (1)$$

(35) According to the above equation, it can be seen that the refractive index of n_2 with a larger value can make the critical angle θ_c larger. That is, less light will be reflected at the interface between the two mediums. Reference is made to FIG. 4. FIG. 4 is a schematic diagram showing two structures, in which the left structure includes the first n-type III-nitride layer **123** and the cathode transparent electrode **130** stacked thereon, and the right structure includes the first n-type III-nitride layer **123**, the second n-type III-nitride layer **124**, and the cathode transparent electrode **130** sequentially stacked. Since the second n-type III-nitride layer **124** of the right structure in FIG. 4 has a larger refractive index than the cathode transparent electrode **130** of the left structure in FIG. 4, the right structure has a larger critical angle θ_c than the left structure. Therefore, compared with the left structure with only the first n-type III-nitride layer **123** and the cathode transparent electrode **130** stacked, the right structure with the first n-type III-nitride layer **123**, the second n-type III-nitride layer **124**, and the cathode transparent electrode **130** can have better light extraction efficiency.

(36) In detail, the second n-type III-nitride layer **124** can be regarded as equivalent to a plurality of layers (for example, approaching infinite layers) stacked sequentially from the top surface **124a** to the bottom surface **124b**, and the refractive index of these layers increases sequentially from the top surface **124a** to the bottom surface **124b**. Since the difference in refractive index any adjacent two of the first n-type III-nitride layer **123**, the layers of the second n-type III-nitride layer **124**, and the cathode transparent electrode **130**, less light will be reflected at the interfaces and thus the light extraction efficiency can be effectively improved.

(37) In some embodiments, the refractive index of the first n-type III-nitride layer **123** may be 2.5. In some embodiments, the refractive index of the second n-type III-nitride layer **124** at the bottom surface **124b** may be equal to or smaller than 2.5. For example, the refractive index of the second n-type III-nitride layer **124** may be in a range from 2.0 to 2.5. Since the difference in refractive index between the first n-type III-nitride layer **123** and the second n-type III-nitride layer **124** is appropriately small, more light propagating in the first n-type III-nitride layer **123** can enter the second n-type III-nitride layer **124** (i.e., less light is reflected at the interface between the first n-type III-nitride layer **123** and the second n-type III-nitride layer **124**). In this way, the light

extraction efficiency of the micro light-emitting diode **120** can be effectively improved.

(38) In some embodiments, the thickness of the second n-type III-nitride layer **124** may be greater than 10 nm. In this way, the second n-type III-nitride layer **124** may have sufficient thickness to appear the effect of refractive index matching.

(39) In some embodiments, the substrate **110** having the anode electrode **111** thereon is a substrate which is capable of driving the micro light-emitting diodes. For example, the substrate **110** has thin-film transistor circuits and thus makes the whole system as a display device.

(40) In some embodiments, the thickness of the second n-type III-nitride layer **124** may be greater than 40 nm. In this way, the second n-type III-nitride layer **124** with the larger thickness can make the effect of refractive index matching appear more significantly. In addition, the flexibility of the processes of manufacturing the micro light-emitting diode **120** can be significantly increased.

(41) In some embodiments, a sum of thicknesses of the p-type III-nitride layer **121**, the active layer **122**, the first n-type III-nitride layer **123**, and the second n-type III-nitride layer **124** is greater than 400 nm. In this way, the micro light-emitting diode **120** can have sufficient strength to cope with subsequent transferring processes.

(42) In some embodiments, a sum of the thicknesses of the first n-type III-nitride layer and the second n-type III-nitride layer is greater than a sum of the thicknesses of the active layer and the p-type III-nitride layer. In this way, the insulation distance between the cathode transparent electrode **130** and the anode electrode **111** can be increased to reduce the possibility of electrical short.

(43) In some embodiments, the sum of the thicknesses of the first n-type III-nitride layer and the second n-type III-nitride layer is greater than 0.5 μm . It makes the isolation layer **140** easier to be thicker to avoid the electrical short.

(44) In some embodiments, the cathode transparent electrode **130** is made of transparent conductive oxide. For example, the transparent conductive oxide may include ITO (Indium Tin Oxide), IZO (indium tin oxide), and AZO (aluminum zinc oxide), but the disclosure is not limited in this regard. The refractive index of the cathode transparent electrode **130** is not greater than the refractive index of the second n-type III-nitride layer **124** at the top surface **124a**.

(45) In some embodiments, the second n-type III-nitride layer **124** is an n-type $\text{Al}_{1-x}\text{Ga}_x\text{N}$ layer. It should be pointed out that “Al” of the n-type $\text{Al}_{1-x}\text{Ga}_x\text{N}$ layer refers to aluminum. “Ga” of the n-type $\text{Al}_{1-x}\text{Ga}_x\text{N}$ layer refers to Gallium. “N” of the n-type $\text{Al}_{1-x}\text{Ga}_x\text{N}$ layer refers to nitrogen. In some embodiments, the Al composition x is equal to or greater than 0.02 and smaller than 1.

(46) In some embodiments, the anode electrode **111** contains metal. In some embodiments, the anode electrode **111** contains Ag—Al alloys, in which “Ag” refers to silver and “Al” refers to aluminum. The Ag—Al alloys may provide a good ohmic contact to the p-type III-nitride layer **121** and good light reflectivity. In some embodiments, the anode electrode **111** containing ITO, IZO, or nickel can also provide a good ohmic contact.

(47) Reference is made to FIG. 5. FIG. 5 is a partial schematic cross-sectional view of a cathode transparent electrode **130** according to some embodiments of the present disclosure. As shown in FIG. 5, in some embodiments, the cathode transparent electrode **130** includes nano metal wires **131** and an encapsulation material **132** in which the nano metal wires **131** are embedded. The nano metal wires **131** are in contact with the second n-type III-nitride layer **124**. The refractive index of the encapsulation material **132** can be regarded as the refractive index of the cathode transparent electrode **130**. The refractive index of the encapsulation material **132** is smaller than the refractive index of the second n-type III-nitride layer **124** at the top surface **124a**.

(48) As shown in FIG. 1, in some embodiments, the micro light-emitting diode device **100A** further includes an isolation layer **140**. The isolation layer **140** is on the substrate **110** and surrounds the micro light-emitting diode **120**. A height H2 of the isolation layer **140** relative to the substrate **110** is equal to or smaller than a height H1 of the top surface **124a** of the second n-type III-nitride layer **124** relative to the substrate **110**. In this way, the contact area between the micro light-emitting

diode **120** and the cathode transparent electrode **130** is larger, so the resistance can be smaller. In addition, the upward light-emitting efficiency of the micro light-emitting diode device **100A** can be better (because the isolation layer **140** does not cover the top surface **124a** of the second n-type III-nitride layer **124**).

(49) In some embodiments, a refractive index of the isolation layer **140** is smaller than the refractive index of the cathode transparent electrode **130**. That is, the difference in refractive index between the first n-type III-nitride layer **123** and the isolation layer **140** will be greater than the difference in refractive index between the first n-type III-nitride layer **123** and the second n-type III-nitride layer **124**. In this way, more light will be reflected at the interface between the first n-type III-nitride layer **123** and the isolation layer **140** than the interface between the first n-type III-nitride layer **123** and the second n-type III-nitride layer **124**, so that the upward light-emitting efficiency of the micro light-emitting diode device **100A** can be further improved.

(50) In some embodiments, the isolation layer **140** includes resin, but the disclosure is not limited in this regard.

(51) Reference is made to FIG. **6**. FIG. **6** is a schematic cross-sectional view of a micro light-emitting diode device **100B** according to some embodiments of the present disclosure. As shown in FIG. **6**, in some embodiments, the micro light-emitting diode device **100B** includes a substrate **110**, a micro light-emitting diode **120**, a cathode transparent electrode **130**, and an isolation layer **140**. A difference between the embodiments shown in FIGS. **1** and **6** is that a height H3 of the isolation layer **140** as shown in FIG. **6** relative to the substrate **110** is greater than the height H1 of the top surface **124a** of the second n-type III-nitride layer **124** relative to the substrate **110**. In this way, the thickness of the isolation layer **140** shown in FIG. **6** is larger (compared to the isolation layer **140** shown in FIG. **1**), which decreases the possibility of electrical short between the cathode transparent electrode **130** and the active layer **122**. Thus, it may benefit the production yield rate of the micro light-emitting diode device **100B**.

(52) Reference is made to FIG. **7A**. FIG. **7A** is a schematic cross-sectional view of a micro light-emitting diode device **100C** according to some embodiments of the present disclosure. As shown in FIG. **7A**, in some embodiments, the micro light-emitting diode device **100C** includes a substrate **110**, a micro light-emitting diode **120**, a cathode transparent electrode **130**, and an isolation layer **140**. A difference between the embodiments shown in FIGS. **1** and **7A** is that the micro light-emitting diode device **100C** as shown in FIG. **7A** further includes an encapsulation layer **150**. The encapsulation layer **150** is above and in contact with the cathode transparent electrode **130**. Specifically, the encapsulation layer **150** encapsulates the micro light-emitting diode **120**, the cathode transparent electrode **130**, and the isolation layer **140**. A refractive index of the encapsulation layer **150** is smaller than the refractive index of the cathode transparent electrode **130**. In this way, more light propagating in the first n-type III-nitride layer **123** can enter the encapsulation layer **150** sequentially via the second n-type III-nitride layer **124** and the cathode transparent electrode **130** with less reflection, such that the light extraction efficiency of the micro light-emitting diode **120** can be further improved.

(53) In some embodiments where the refractive index of the second n-type III-nitride layer **124** is in a range from 2.0 to 2.5, the refractive index of the encapsulation layer **150** is smaller than 2.0.

(54) Reference is made to FIG. **7B**. FIG. **7B** is a schematic cross-sectional view of a micro light-emitting diode device **100C-1** according to some embodiments of the present disclosure. As shown in FIG. **7B**, in some embodiments, the micro light-emitting diode device **100C-1** includes a substrate **110**, a micro light-emitting diode **120**, a cathode transparent electrode **130**, an isolation layer **140**, and an encapsulation layer **150a**. A difference between the embodiments shown in FIGS. **7A** and **7B** is that the encapsulation layer **150** as shown in FIG. **7A** has a bulb shape, while the encapsulation layer **150a** as shown in FIG. **7B** has a layered shape. The encapsulation layer **150a** is above and in contact with the cathode transparent electrode **130**. Specifically, the encapsulation layer **150a** encapsulates the micro light-emitting diode **120**, the cathode transparent electrode **130**,

and the isolation layer **140**. Another difference between the embodiments shown in FIGS. 7A and 7B is that the micro light-emitting diode device **100C-1** as shown in FIG. 7B further includes a covering film **180**. The covering film **180** covers and is in contact with a surface of the encapsulation layer **150a** away from the substrate **110**. A refractive index of the encapsulation layer **150a** is smaller than the refractive index of the cathode transparent electrode **130**. In this way, more light propagating in the first n-type III-nitride layer **123** can enter the encapsulation layer **150a** sequentially via the second n-type III-nitride layer **124** and the cathode transparent electrode **130** with less reflection, such that the light extraction efficiency of the micro light-emitting diode **120** can be further improved.

(55) In some embodiments, as shown in FIG. 7B, the encapsulation layer **150a** includes inorganic particles **150b** evenly distributed therein. In this way, the refractive index of the encapsulation layer **150a** can be adjusted by the inorganic particles **150b**. In some embodiments, the inorganic particles **150b** are TiO₂ particles, but the disclosure is not limited in this regard.

(56) Reference is made to FIG. 8. FIG. 8 is a schematic cross-sectional view of a micro light-emitting diode device **100D** according to some embodiments of the present disclosure. As shown in FIG. 8, in some embodiments, the micro light-emitting diode device **100D** includes a substrate **110**, a micro light-emitting diode **120**, a cathode transparent electrode **130**, and an isolation layer **140**. A difference between the embodiments shown in FIGS. 1 and 8 is that the micro light-emitting diode device **100D** as shown in FIG. 8 further includes a dielectric layer **160**. The dielectric layer **160** is between the isolation layer **140** and a side surface of the micro light-emitting diode **120**. The dielectric layer **160** can reduce surface recombination on the side surface of the micro light-emitting diode **120** and thus it may improve the radiative recombination rate of the micro light-emitting diode **120**.

(57) Reference is made to FIG. 9. FIG. 9 is a schematic cross-sectional view of a micro light-emitting diode device **100E** according to some embodiments of the present disclosure. As shown in FIG. 9, in some embodiments, the micro light-emitting diode device **100E** includes a substrate **110**, a micro light-emitting diode **120**, a cathode transparent electrode **130**, and an isolation layer **140**. A difference between the embodiments shown in FIGS. 1 and 9 is that the micro light-emitting diode device **100E** further includes an MgO layer **170**. The MgO layer **170** covers and is in contact with the cathode transparent electrode **130**. A refractive index of the MgO layer **170** is smaller than the refractive index of the cathode transparent electrode **130**. In this way, more light propagating in the first n-type III-nitride layer **123** can enter the MgO layer **170** sequentially via the second n-type III-nitride layer **124** and the cathode transparent electrode **130** with less reflection, such that the light extraction efficiency of the micro light-emitting diode **120** can be further improved.

(58) Reference is made to FIG. 10. FIG. 10 is a schematic cross-sectional view of a micro light-emitting diode device **100F** according to some embodiments of the present disclosure. As shown in FIG. 10, in some embodiments, the micro light-emitting diode device **100F** includes a substrate **110**, a micro light-emitting diode **120**, a cathode transparent electrode **130**, an isolation layer **140**, and an MgO layer **170**. A difference between the embodiments shown in FIGS. 9 and 10 is that the cathode transparent electrode **130** and the MgO layer **170** of the micro light-emitting diode device **100F** have larger coverage areas. In this way, the light output distribution of the micro light-emitting diode device **100F** can be more symmetric.

(59) Reference is made to FIG. 11. FIG. 11 is a schematic top view of a micro light-emitting diode device **100G** according to some embodiments of the present disclosure. As shown in FIG. 11, the micro light-emitting diode device **100G** includes a substrate **110**, a plurality of micro light-emitting diodes **120R**, **120G**, **120B**, and an isolation layer **140**. The micro light-emitting diodes **120R**, **120G**, **120B** are disposed on the substrate **110**. Each of the micro light-emitting diodes **120R**, **120G**, **120B** has a structure similar to that of the micro light-emitting diode **120** shown in FIG. 1 and has a lateral width smaller than 100 μm. The isolation layer **140** is disposed on the substrate **110** and surrounds the micro light-emitting diodes **120R**, **120G**, **120B**. The micro light-emitting diodes

120R, 120G, 120B are configured to emit lights of different wavelengths respectively. For example, the micro light-emitting diodes **120R, 120G, 120B** are configured to emit a red light, a green light, and a blue light. Since the micro light-emitting diodes **120R, 120G, 120B** are surrounded by the same isolation layer **140**, lights emitted by the micro light-emitting diodes **120R, 120G, 120B** can be evenly mixed.

(60) In some embodiments, a distance between any two of the micro light-emitting diodes **120R, 120G, 120B** is less than 500 μm . In this way, lights emitted by the micro light-emitting diodes **120R, 120G, 120B** can be mixed more evenly.

(61) Reference is made to FIG. **12**. FIG. **12** is a schematic top view of a micro light-emitting diode device **100H** according to some embodiments of the present disclosure. Compared to the micro light-emitting diode device **100G** shown in FIG. **11**, the micro light-emitting diode device **100H** further includes a micro light-emitting diode **120Y**. The micro light-emitting diode **120Y** is also surrounded by the isolation layer **140**, and is configured to emit a yellow light. Since the micro light-emitting diodes **120R, 120G, 120B, 120Y** are surrounded by the same isolation layer **140**, lights emitted by the micro light-emitting diodes **120R, 120G, 120B, 120Y** can be evenly mixed. In some embodiments, a distance of any two of the micro light-emitting diodes **120R, 120G, 120B, 120Y** is less than 2 mm. In this way, lights emitted by the micro light-emitting diodes **120R, 120G, 120B, 120Y** can be mixed more evenly.

(62) According to the foregoing recitations of the embodiments of the disclosure, it can be seen that in the micro light-emitting diode device of the disclosure, the micro light-emitting diode includes the second n-type III-nitride layer that contains aluminum relative to the first n-type III-nitride layer, so the refractive index of the second n-type III-nitride layer with the wider band gap will be smaller than the refractive index of the first n-type III-nitride layer with the narrower band gap, which can effectively improve the light extraction efficiency of the micro light-emitting diode. In addition, since the second n-type III-nitride layer has a refractive index varying in a monotonically non-decreasing manner from the top surface thereof, the light extraction efficiency can be further improved.

(63) Although the present disclosure has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

(64) It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the present disclosure cover modifications and variations of this disclosure provided they fall within the scope of the following claims.

Claims

1. A micro light-emitting diode device, comprising: a substrate having an anode electrode thereon; a micro light-emitting diode having a lateral width smaller than 100 μm , disposed on and in contact with the anode electrode, and comprising: a p-type III-nitride layer; a first n-type III-nitride layer above the p-type III-nitride layer; a second n-type III-nitride layer above and in contact with the first n-type III-nitride layer, wherein the second n-type III-nitride layer contains aluminum and has a top surface and a bottom surface respectively away from and adjacent to the first n-type III-nitride layer, a refractive index of the second n-type III-nitride layer is smaller than a refractive index of the first n-type III-nitride layer and varies in a monotonically non-decreasing manner from the top surface, and the refractive index of the second n-type III-nitride layer is larger at the bottom surface than at the top surface; and an active layer between the p-type III-nitride layer and the first n-type III-nitride layer; and a cathode transparent electrode at least partially in contact with the top surface of the second n-type III-nitride layer.

2. The micro light-emitting diode device of claim 1, wherein the refractive index of the second n-

type III-nitride layer monotonically increase from the top surface.

3. The micro light-emitting diode device of claim 1, further comprising: an isolation layer on the substrate and surrounding the micro light-emitting diode, wherein a height of the isolation layer relative to the substrate is equal to or smaller than a height of the top surface of the second n-type III-nitride layer relative to the substrate.

4. The micro light-emitting diode device of claim 3, wherein a refractive index of the isolation layer is smaller than a refractive index of the cathode transparent electrode.

5. The micro light-emitting diode device of claim 3, further comprising: a dielectric layer between the isolation layer and a side surface of the micro light-emitting diode.

6. The micro light-emitting diode device of claim 1, wherein the cathode transparent electrode is made of transparent conductive oxide, and a refractive index of the cathode transparent electrode is smaller than the refractive index of the second n-type III-nitride layer.

7. The micro light-emitting diode device of claim 1, further comprising an encapsulation layer above and in contact with the cathode transparent electrode, wherein the cathode transparent electrode comprises nano metal wires in contact with the second n-type III-nitride layer.

8. The micro light-emitting diode device of claim 1, further comprising: an encapsulation layer above and in contact with the cathode transparent electrode, wherein a refractive index of the encapsulation layer is smaller than a refractive index of the cathode transparent electrode.

9. The micro light-emitting diode device of claim 8, wherein the refractive index of the encapsulation layer is smaller than 2.0.

10. The micro light-emitting diode device of claim 8, further comprising: an isolation layer on the substrate and surrounding the micro light-emitting diode, wherein the refractive index of the encapsulation layer is greater than a refractive index of the isolation layer.

11. The micro light-emitting diode device of claim 1, further comprising: an MgO layer covering and in contact with the cathode transparent electrode, wherein a refractive index of the MgO layer is smaller than a refractive index of the cathode transparent electrode.

12. The micro light-emitting diode device of claim 1, further comprising: an isolation layer on the substrate and surrounding the micro light-emitting diode, wherein a height of the isolation layer relative to the substrate is greater than a height of the top surface of the second n-type III-nitride layer relative to the substrate.

13. The micro light-emitting diode device of claim 12, wherein a refractive index of the isolation layer is smaller than the refractive index of the cathode transparent electrode.

14. The micro light-emitting diode device of claim 12, further comprising: a dielectric layer between the isolation layer and a side surface of the micro light-emitting diode.

15. The micro light-emitting diode device of claim 1, wherein the refractive index of the second n-type III-nitride layer is equal to or smaller than 2.5.

16. The micro light-emitting diode device of claim 1, wherein an atomic ratio of aluminum in the second n-type III-nitride layer varies in a monotonically non-increasing manner from the top surface.

17. The micro light-emitting diode device of claim 16, wherein the atomic ratio of aluminum in the second n-type III-nitride layer monotonically decreases from the top surface.

18. The micro light-emitting diode device of claim 1, wherein the second n-type III-nitride layer is an n-type $\text{Al}_{\text{sub}.x}\text{Ga}_{\text{sub}.1-x}\text{N}$ layer.

19. The micro light-emitting diode device of claim 1, wherein a thickness of the second n-type III-nitride layer is greater than 40 nm.

20. The micro light-emitting diode device of claim 1, wherein the anode electrode contains metal.

21. The micro light-emitting diode device of claim 20, wherein the anode electrode contains Ag—Al alloys.

22. The micro light-emitting diode device of claim 20, wherein the anode electrode contains nickel.

23. The micro light-emitting diode device of claim 1, wherein a sum of thicknesses of the p-type

III-nitride layer, the active layer, the first n-type III-nitride layer, and the second n-type III-nitride layer is greater than 400 nm.

24. The micro light-emitting diode device of claim 1, wherein a sum of thicknesses of the first n-type III-nitride layer and the second n-type III-nitride layer is greater than a sum of thicknesses of the active layer and the p-type III-nitride layer.

25. The micro light-emitting diode device of claim 1, wherein a sum of thicknesses of the first n-type III-nitride layer and the second n-type III-nitride layer is greater than 0.5 μm .

26. The micro light-emitting diode device of claim 1, further comprising: another micro light-emitting diode disposed on the substrate, wherein the micro light-emitting diode and the another micro light-emitting diode are configured to emit lights of different wavelengths respectively; and an isolation layer disposed on the substrate and surrounding the micro light-emitting diode and the another micro light-emitting diode.

27. The micro light-emitting diode device of claim 26, wherein a distance between the micro light-emitting diode and the another micro light-emitting diode is less than 500 μm .
